





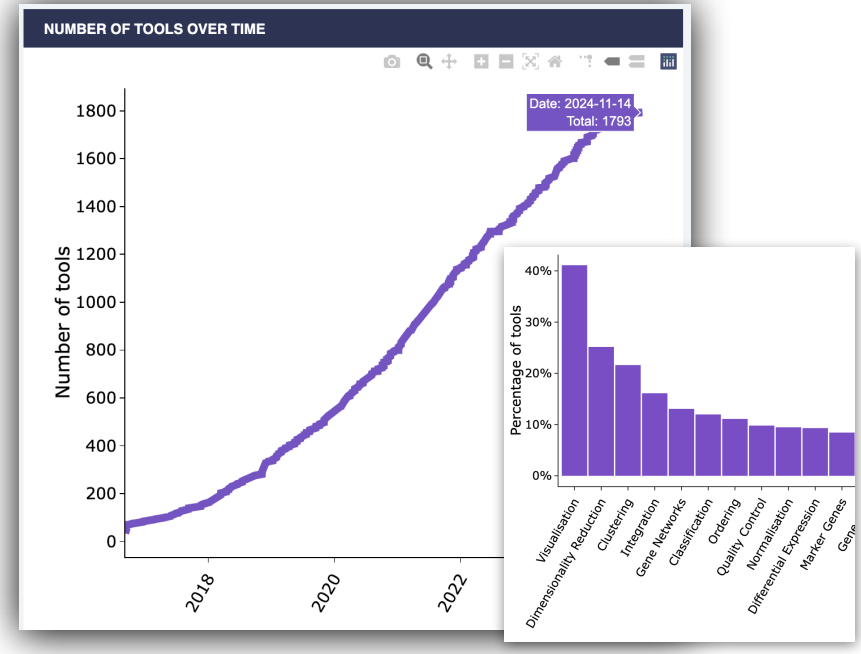
Constructing standardized benchmarks using Omnibenchmark

Mark D. Robinson (credit also to: Ben Carrillo)

What is the situation in computational biology?
Explosion of computational methods! What should I use?

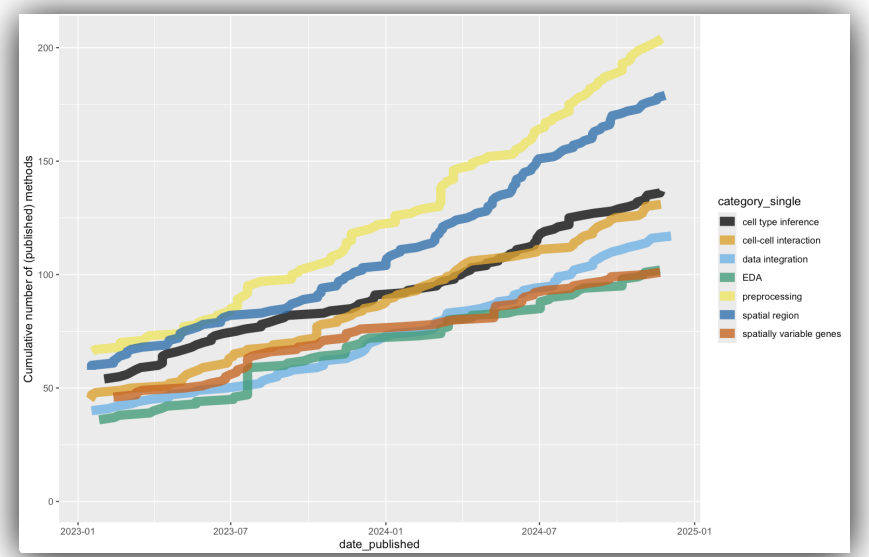
Exploring the single-cell RNA-seq analysis landscape with the scRNA-tools database

Luke Zappia, Belinda Phipson, Alicia Oshlack



Museum of spatial transcriptomics

Lambda Moses[1] and Lior Pachter[1,2]



A common tension in **method evaluation** is ..



CORRESPONDENCE

The self-assessment trap: can we all be better than average?

Raquel Norel, John Jeremy Rice and Gustavo Stolovitzky
IBM Computational Biology Center, IBM T.J. Watson Research Center,
Yorktown Heights, NY, USA

"*researchers wishing to publish their analytical methods are required by referees to compare the performance of their own algorithms against other methodologies, thus being forced to be judge, jury and executioner. The result is that the authors' method tends to be the best ..*" (from 2011!)

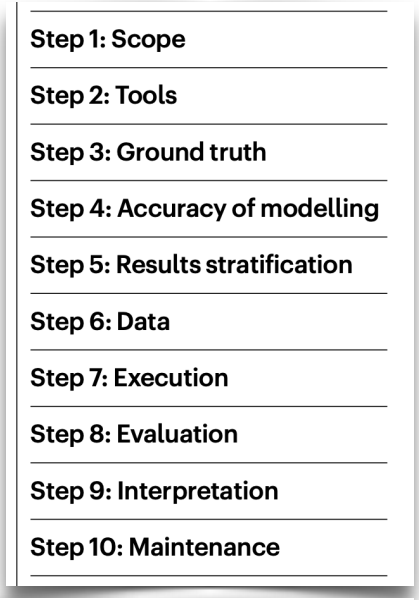
→ we need neutral benchmarking

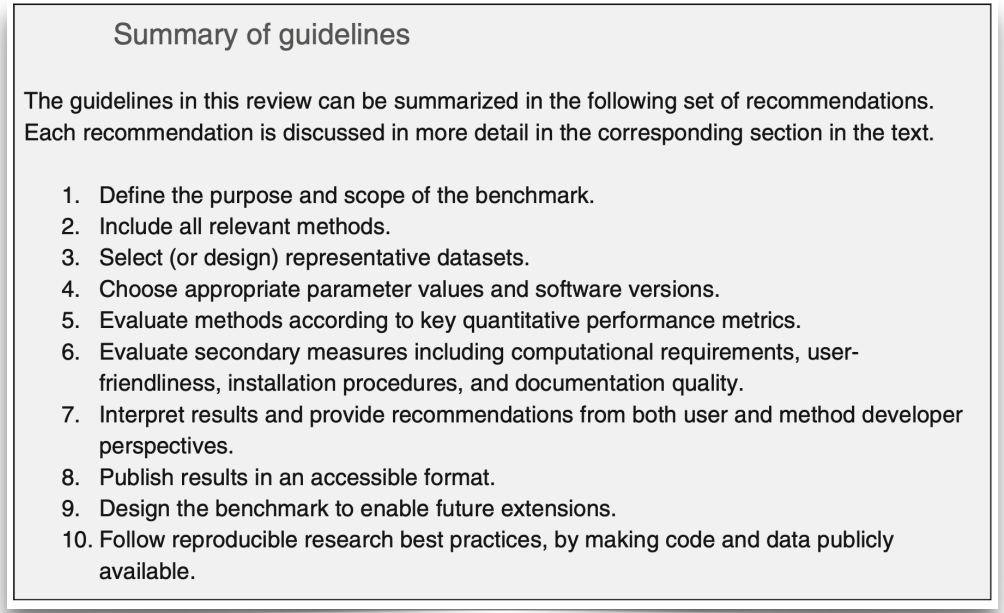
What is a benchmark? And all the issues that come with them ..

- **Task:** e.g., variant calling, cell-type annotation, spot deconvolution, differential expression
- **Datasets:** real data with orthogonal ground truth, (semi-)simulations, or spike-ins
- **Methods:** the tools compared, with documented versions and parameter settings
- **Metrics:** accuracy-type measures plus runtime, memory, scalability, usability
- ***Execution:*** *ideally a reproducible workflow, so it can be scrutinized, rerun and / or extended*

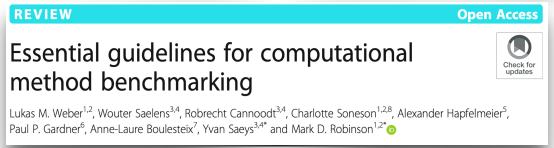
- Other modes: community events, challenges (e.g., **https://virtualcellchallenge.org/**)

How to benchmark? (Today: mostly the technical side)

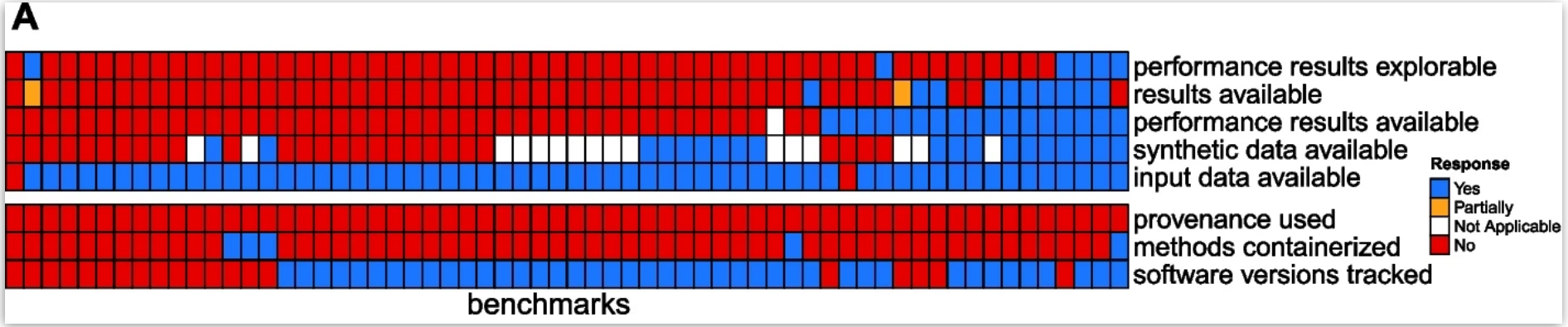








What were 2023 (single-cell method) benchmarks like?

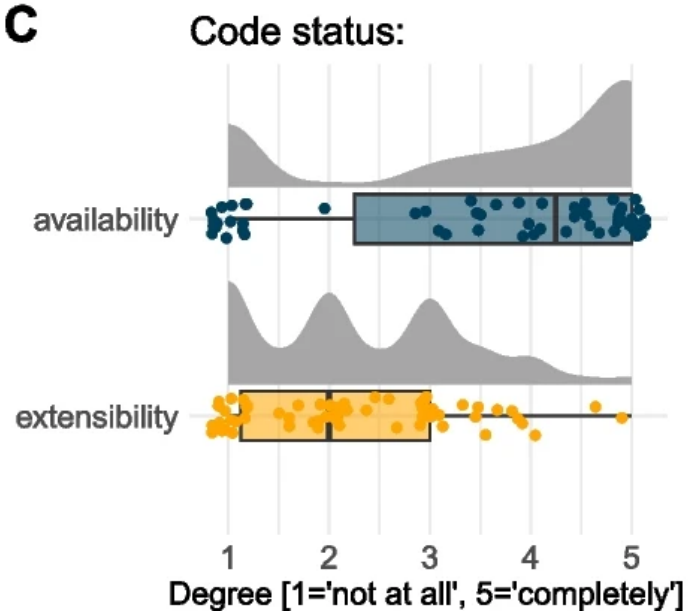


CORRESPONDENCE Open Access

Meta-analysis of (single-cell method) benchmarks reveals the need for extensibility and interoperability

Anthony Sonrel[1,2†], Almut Luetge[1,2†], Charlotte Soneson[2,3†], Izaskun Mallona[1,2,4†], Pierre-Luc Germain[1,2,5†], Sergey Knyazev[6,7], Jeroen Gilis[8,9,10], Reto Gerber[1,2], Ruth Seurinck[8,9], Dominique Paul[1], Emanuel Sonder[1,2,5], Helena L. Crowell[1,2], Imran Fanaswala[1,2], Ahmad Al-Ajami[1,2], Elyas Heidari[1,2], Stephan Schmeing[1,2], Stefan Milosavljevic[1,2,11], Yvan Saeys[8,9], Serghei Mangul[7] and Mark D. Robinson[1,2*]

What would an ideal benchmarking ecosystem contain?



Themes for Omnibenchmark:

- transparency builds trust
- many useful benchmarking artifacts
- benchmarks need extensibility
- benchmarks need standardization → interoperability

PERSPECTIVE

Building a continuous benchmarking ecosystem in bioinformatics

Izaskun Mallona[1,2*], Charlotte Soneson[2,3], Ben Carrillo[1,2], Almut Lütge[1,2,4], Daniel Incicau[1], Reto Gerber[1,2], Anthony Sonrel[1,2], Mark D. Robinson[1,2*]

1 Department of Molecular Life Sciences, University of Zurich, Zurich, Switzerland, **2** SIB Swiss Institute of Bioinformatics, University of Zurich, Zurich, Switzerland, **3** Friedrich Miescher Institute for Biomedical Research, Basel, Switzerland, **4** Swiss Data Science Centre, Zurich, Switzerland

* izaskun.mallona@gmail.com (IM); mark.robinson@mls.uzh.ch (MDR)

Omnibenchmark: open modular benchmarks

Omnibenchmark: transparent, reproducible, extensible and standardized orchestration of solo and collaborative benchmarks

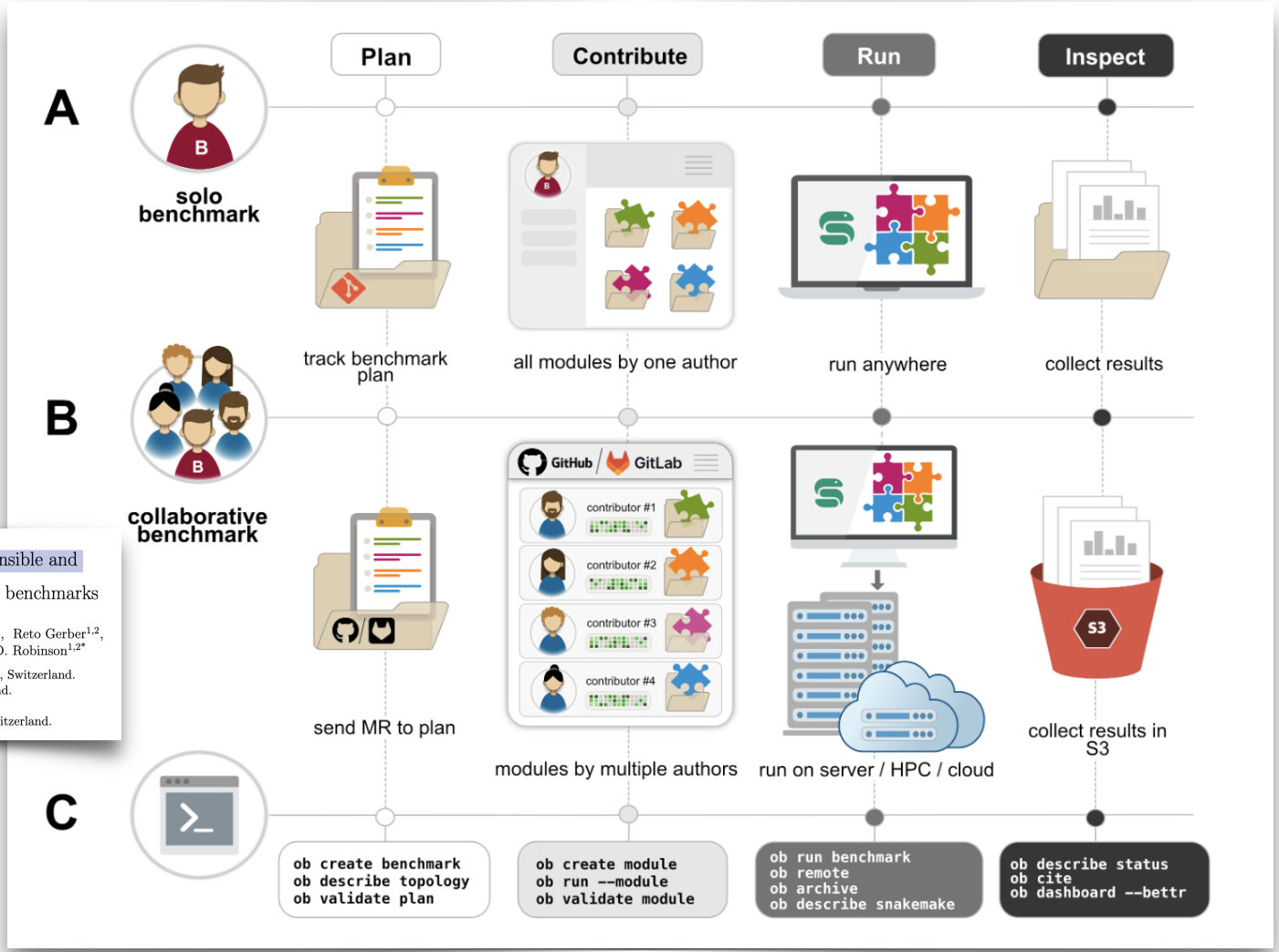
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[1]Department of Molecular Life Sciences, University of Zurich, Zurich, Switzerland.
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arxiv: 2409.17038

Latest update: Aug 12th 2026, accepted Oct 2nd 2026

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What does Omnibenchmark do?

- Takes a “benchmark plan” in YAML format
- Sketches a (Snakemake) workflow of the benchmark
- Generates *standardized* “benchmark artifacts” (software envs, datasets, methods run on datasets, performance results), which can be shared, inspected, and analyzed
- Omnibenchmark guarantees*: reproducibility, transparency, extensibility
- Omnibenchmark *does not* guarantee: good science (e.g., realistic simulations, appropriate metrics, use)

The Omnibenchmark philosophy

- Lock down
  - software environments
  - code (via repo hash)
  - inputs / outputs
- Generate standardized (archivable) outputs
- Facilitates open collaborative benchmarking

Longer term: next frontier is benchmarking the AIs

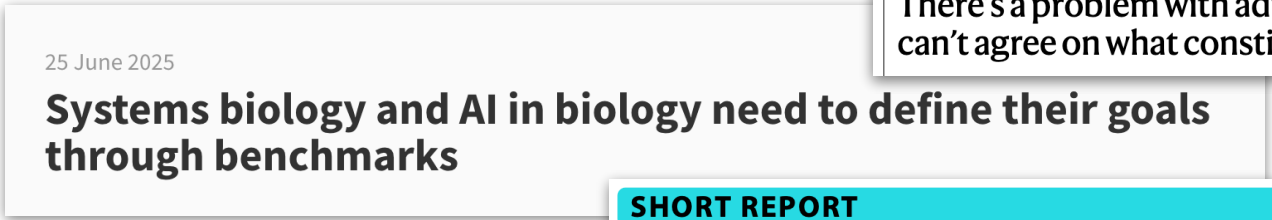
Anshul Kundaje sums up his frustration with the use of artificial intelligence in science in three words: "**bad benchmarks propagate**".


Brief Communication https://doi.org/10.1038/s41592-025-02772-6

Deep-learning-based gene perturbation effect prediction does not yet outperform simple linear baselines


BUILDING BETTER BENCHMARKS

There's a problem with advancing AI for science: researchers can't agree on what constitutes progress. **By Michael Brooks**


25 June 2025

Systems biology and AI in biology need to define their goals through benchmarks


SHORT REPORT

Zero-shot evaluation reveals limitations of single-cell foundation models

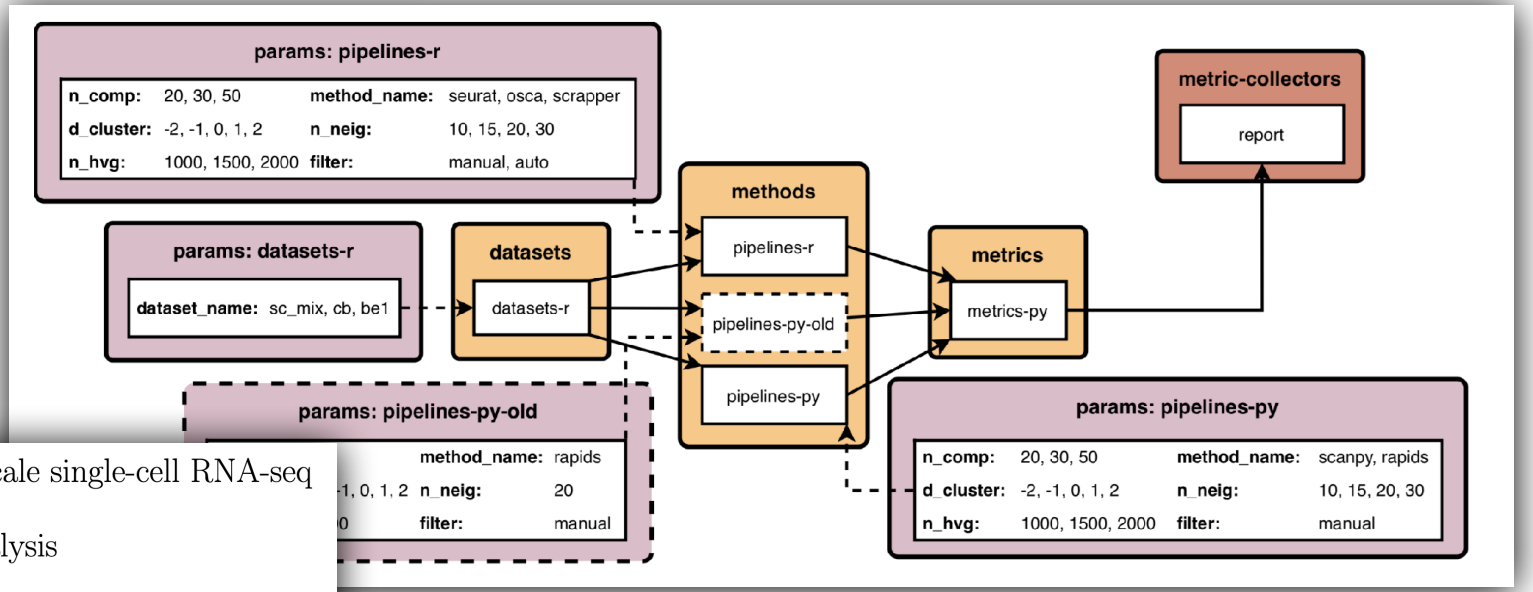
Kasia Z. Kedzierska[1], Lorin Crawford[2], Ava P. Amini[2] and Alex X. Lu[2*]

Omnibenchmark: open modular benchmarks

Building computational benchmarks: an Omnibenchmark reimplementation of a single-cell preprocessing pipeline evaluation

Atreya Choudhury[1,2], Tom Kitak[1,2], Ben Carrillo[1,2], Pauline Busch[1,2], Martin Emons[1,2], Samuel Gunz[1,2], Maruša Koderman[1,2], Siyuan Luo[1,2], Izaskun Mallona[1,2], Aidan Meara[1], David Wissel[1,2], Mark D. Robinson[1,2*]

[1]Department of Molecular Life Sciences, University of Zürich, Winterthurerstrasse 190, Zürich, 8057, Zurich, Switzerland.
[2]Swiss Institute of Bioinformatics, University of Zurich, Winterthurerstrasse 190, Zurich, 8057, Zurich, Switzerland.



Motivating benchmark:

Benchmarking large-scale single-cell RNA-seq analysis

Ilaria Billato[1], Herve Pages[2], Vince Carey[3], Levi Waldron[4], Gabriele Sales[1†], Chiara Romualdi[1†], Davide Risso[5*†]

Single cell data analysis pipelines: our (motivating) example benchmark

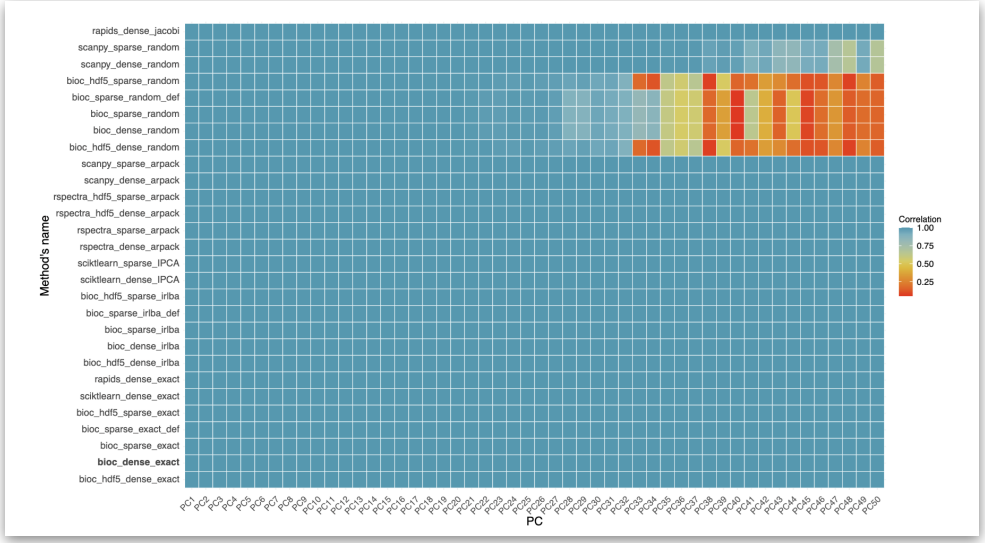
Benchmarking large-scale single-cell RNA-seq

analysis

Ilaria Billato[1], Herve Pages[2], Vince Carey[3], Levi Waldron[4], Gabriele Sales[1†], Chiara Romualdi[1†], Davide Risso[5*†]

ARI	Algorithm	Seurat	OSCA	Scanpy	Rapids	scrapper
cb	louvain	0.88	0.81	0.75	0.78	0.80
	leiden	0.7	0.81	0.75	0.77	0.81
sc_mix	louvain	0.97	0.96	0.99	0.99	0.96
	leiden	0.98	0.96	0.99	0.99	0.96
BE1	louvain	0.67	0.97	0.68	0.68	0.97
	leiden	0.68	0.95	0.68	0.68	0.97

Table 1 Adjusted Rand Index (ARI) between the workflow and between single-cell RNA-seq dataset.



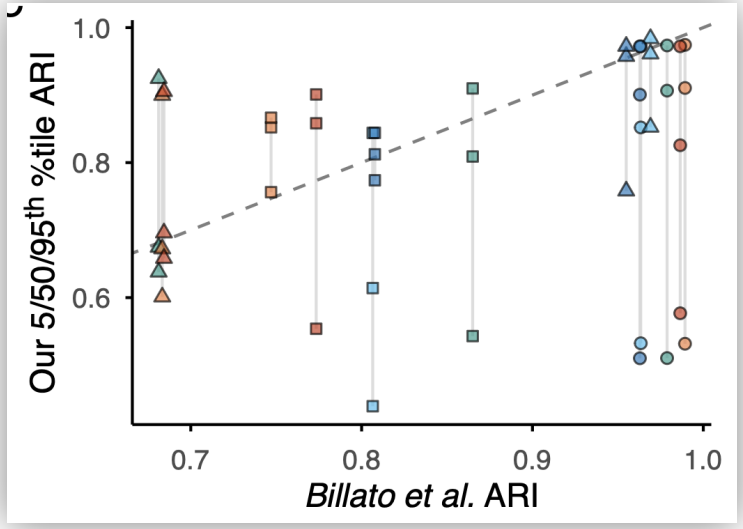
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Benchmarking large-scale single-cell RNA-seq analysis

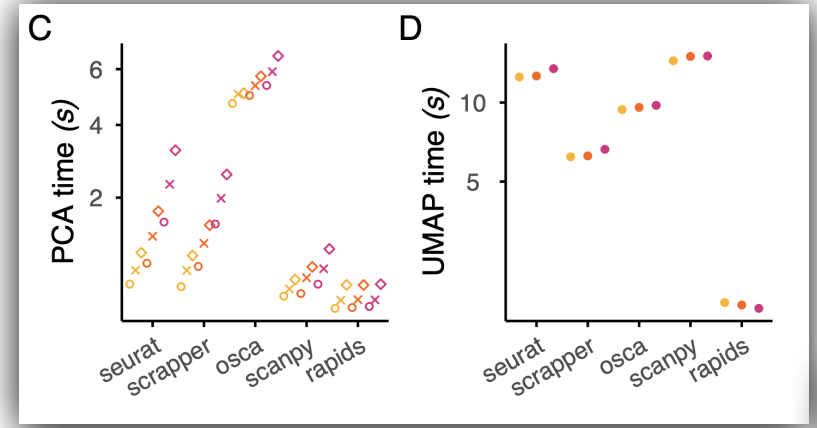
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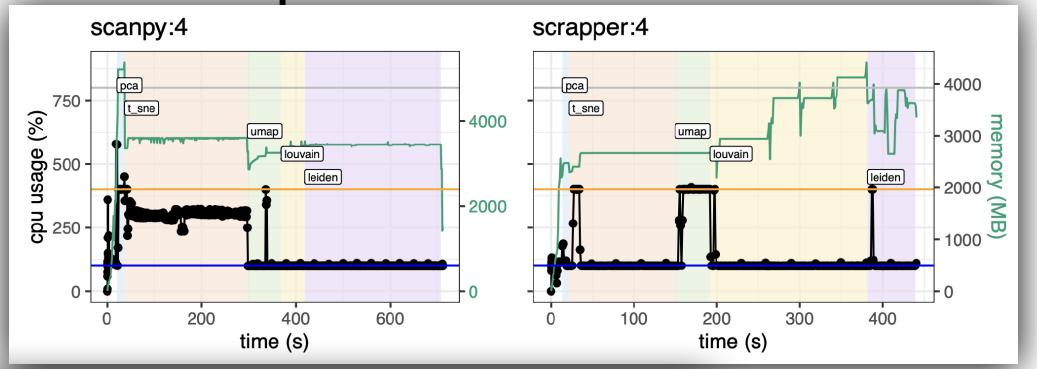
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	leiden	0.98	0.96	0.99	0.99	0.96
BE1	louvain	0.67	0.97	0.68	0.68	0.97
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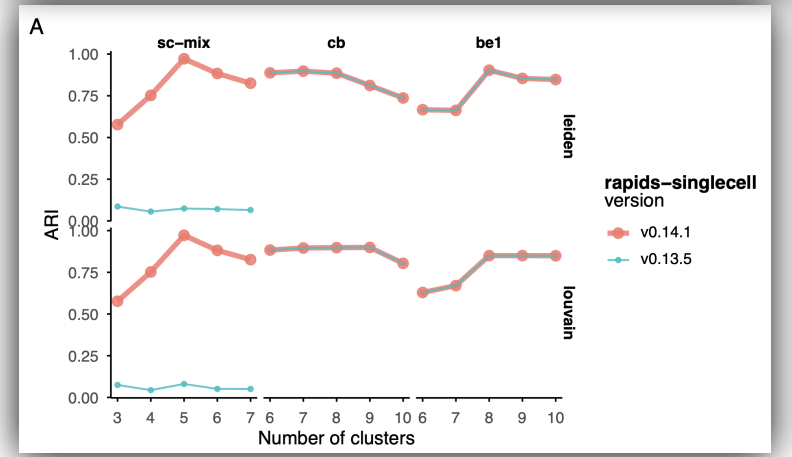
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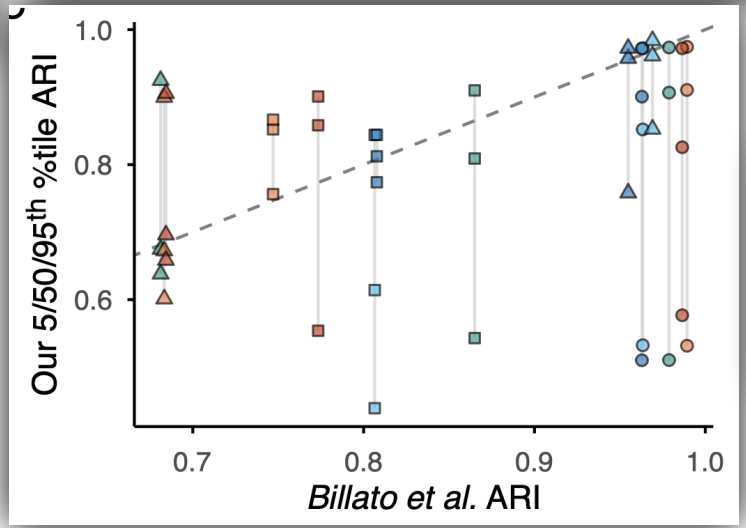


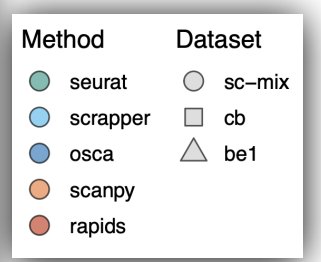
Single cell pipelines: performance











Omnibenchmark: under the hood

```
name: scRNA Benchmark (pipelines)
id: pipelines-plan
description: "Benchmarking scRNA pipelines (as implemented by Billato et al.: 10.1101/2025.10.28.681564)"
benchmarker: Atreya Choudhury, Mark Robinson, Martin Emons, David Wissel, Samuel Gunz
version: "1.1"

software_backend: apptainer
software_environments:
  r_apptainer:
    name: R apptainer
    apptainer: oras://quay.io/omnibenchmark/scrna-bench:r-bioc-320-latest-amd64
  py_apptainer_rsc_0.13.5:
    name: python apptainer with rsc v0.13.5
    apptainer: oras://quay.io/omnibenchmark/scrna-bench:py-rsc-
  py_apptainer_rsc_0.14.1:
    name: python apptainer with rsc v0.14.1
    apptainer: oras://quay.io/omnibenchmark/scrna-bench:py-rsc-

stages:
  - id: datasets
    outputs:
      - id: data.h5ad
        path: "{dataset}.h5ad"
      - id: data.clusters_truth
        path: "{dataset}.clusters_truth.tsv"
      - id: data.clusters_truth_num
        path: "{dataset}.clusters_truth_num.txt"
    modules:
      - id: datasets-r
        name: loads scRNA datasets
        software_environment: r_apptainer
        repository:
          url: https://github.com/scrna-bench/datasets
          commit: "d2e05fc"
        parameters:
          - dataset_name: ["pbmc68k"]
          # - dataset_name: ["sc-mix", "be1", "cb", "1.3m"]
```

“Self-describing” plan

```
(base) mark@MLS-M-MAROBI pipelines-plan % ll out/.modules
total 0
drwxr-xr-x  8 mark  staff  256 Apr 25 14:04 datasets
drwxr-xr-x  3 mark  staff   96 Apr 25 14:04 metrics
drwxr-xr-x  7 mark  staff  224 Apr 25 14:04 pipelines-py
drwxr-xr-x  4 mark  staff  128 Apr 25 14:04 pipelines-r
drwxr-xr-x  3 mark  staff   96 Apr 24 17:10 pipelines-report
(base) mark@MLS-M-MAROBI pipelines-plan % ll out/datasets
total 0
drwxr-xr-x  4 mark  staff  128 Apr 24 09:56 datasets-r
(base) mark@MLS-M-MAROBI pipelines-plan % ll out/.logs
total 1008
-rw-r--r--  1 mark  staff      0 Apr 24 10:06 collector_report_default.log
-rw-r--r--  1 mark  staff   4757 Apr 24 09:56 datasets_datasets_r_8b2abb9e.log
-rw-r--r--  1 mark  staff   4618 Apr 25 15:07 datasets_datasets_r_8b2abb9e_methods_pipelines_py_be
-rw-r--r--  1 mark  staff    139 Apr 25 14:14 datasets_datasets_r_8b2abb9e_methods_pipelines_py_be
-rw-r--r--  1 mark  staff   4617 Apr 25 15:18 datasets_datasets_r_8b2abb9e_methods_pipelines_py_de
-rw-r--r--  1 mark  staff    139 Apr 25 14:16 datasets_datasets_r_8b2abb9e_methods_pipelines_py_de
-rw-r--r--  1 mark  staff   4615 Apr 25 15:37 datasets_datasets_r_8b2abb9e_methods_pipelines_py_f8
-rw-r--r--  1 mark  staff    139 Apr 25 14:24 datasets_datasets_r_8b2abb9e_methods_pipelines_py_f8
-rw-r--r--  1 mark  staff  10446 Apr 25 15:53 datasets_datasets_r_8b2abb9e_methods_pipelines_r_340
```

Code modules

Benchmark outputs

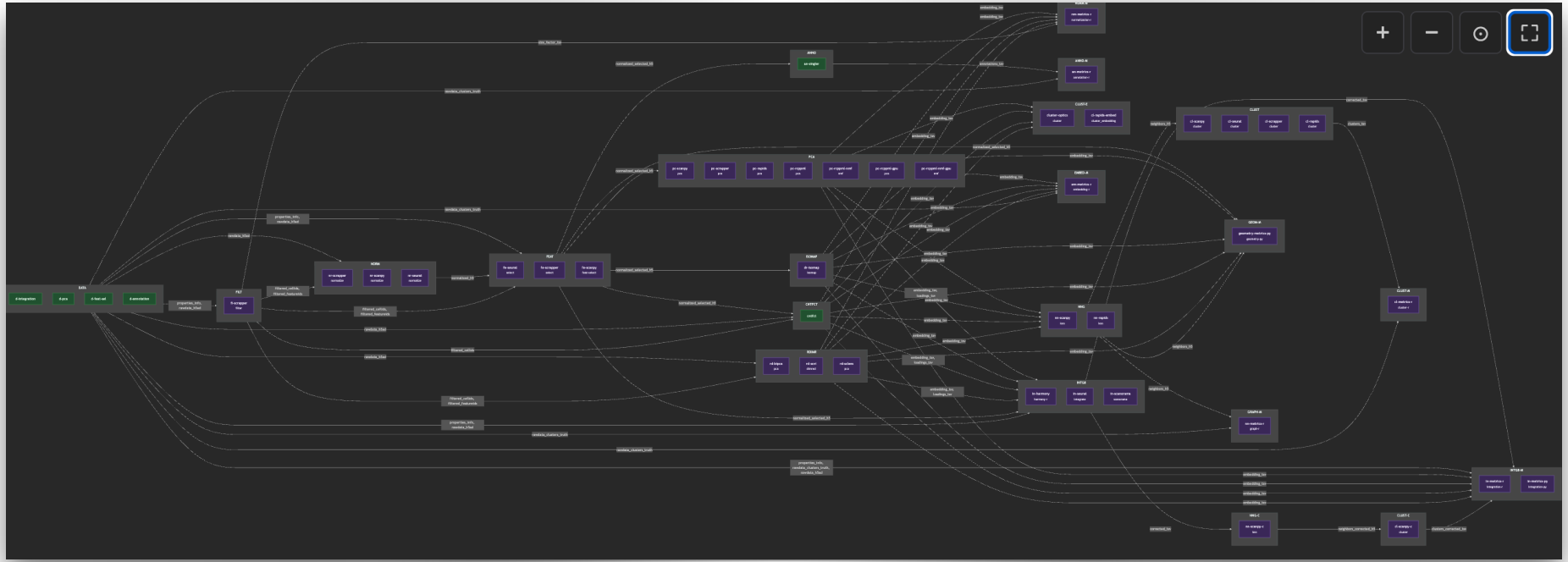
Logs

Standardized outputs

Omnibenchmark: where is this going?

i) interoperable benchmarking modules

ii) Benchmark scaffold (large graph of many possible pipeline connections) —> slices



Demo

Step 0: Find a Terminal; Install conda

Step 1: Create a conda environment based on Python 3.12 and install Omnibenchmark into it (see `README.md` at plan repo: **https://github.com/sta426hs2026/omni-pipelines-plan**)

Step 2: Validate and run the `omni-pipelines-plan` plan (see same `README.md`)

(while Step 2 is running)

Step 3: Explore:

- plan
- conda environments
- obeditor: **https://omnibenchmark.github.io/obeditor/**
- `out/` directory
- Snakefile
- logs
- outputs